Supplementary Materials for the Paper: SALE-MLP: Structure Aware Latent Embeddings for GNN to Graph-free MLP Distillation

1 Datasets

The statistical analysis of the following six public benchmark datasets used during experimentation: Cora[Sen *et al.*, 2008], Citeseer[Sen *et al.*, 2008], Pubmed[Sen *et al.*, 2008], Amazon-Photo[Feng *et al.*, 2022], Amazon-Computer[Feng *et al.*, 2022] and large-scale graph ogbn-arxiv[Hu *et al.*, 2020] is presented in Table 1

Table 1: Overview of Datasets

Metric	**Cora**	**Citeseer**	**Pubmed**	**A-computer**	**A-photo**	**Ogbn-arxiv**
# Nodes	2,485	2,110	19,717	13,381	7,487	169,343
# Edges	5,069	3,668	44,324	245,778	119,043	1,166,243
# Features	1,433	3,703	500	767	745	128
# Classes	7	6	3	10	8	40
% Heterophily	15.73	20.47	16.17	21.69	16.75	32.16
Mean Degree	5.08	4.48	5.50	37.74	32.80	14.77

2 Model Hyperparameters

The hyperparameters used to train SALE-MLP for different datasets after using grid search are reported in Table 2.

Table 2: Hyperparameters for SALE-MLP

	Cora	**Citeseer**	**Pubmed**	**A-Photo**	**A-Computer**	**Ogn-arxiv**
dim	128	128	128	128	128	256
# layers	2	2	2	2	2	3
# walks	3	3	3	2	2	5
Walk len	20	20	20	10	10	50
λ	0.6	0	0.5	0.1	0	0
α	2.5	2.5	2.5	0.001	0.001	0.0001
pre Ep	10	10	10	10	10	2
total Ep	200	200	200	200	200	500

Moreover, Tab. 3 shows the impact of window size, showing the accuracy and the time per iteration. For Cora and Citeseer, increasing the window size slightly increases the training time as more information needs to be aggregated. Further, the performance (%) peaks around 10-15 window size (above and below which performance drops). This can be due to the fact that as we increase the window size, we move further away from the node, which can lead to using less important information

Table 3: Analysis of DeepWalk window size (Acc. (%) / time (sec)).

Window Size	5	10	15	20
Cora	83.29%/1.17s	83.73%/1.20s	82.83%/1.21s	82.80%/1.23s
Citeseer	72.34%/1.32s	73.17%/1.33s	73.44%/1.34s	73.10%/1.36s

3 Time and Space Complexity

The time of any GNN is dominated by message passing: O(N.k.d). Here, N: number of nodes, k: neighbor count, and d: feature dimension. It also requires the line transformation, giving additional complexity of O(N.d.d1), where, d1 is the hidden dimension. Total complexity: O(N.k.d + N.d.d1). On the other hand, SALE-MLP just needs to compute the linear transformation, leading to complexity: O(N.d.d1). In large real-world graphs, k is huge as the number of neighbors increases. Lastly, for space complexity, the GNN-based method needs to store graph during inference time, while SALE-MLP is graph-free.

4 Design Choices

4.1 Loss Agnostic

The proven performance of structural loss on large-scale datasets motivates us to propose loss loss-agnostic method SALE-MLP which can be trained using any available unsupervised structural loss. To establish we have experimented on various unsupervised structural losses like Line[Tang *et al.*, 2015], DeepWalk[Perozzi *et al.*, 2014], and Node2vec[Grover and Leskovec, 2016].

Table 4: Effect of different unsupervised structural losses

Structural Loss	**Cora**	**Citeseer**
SAGE	81.03	69.14
SALE-MLP(Deepwalk)	83.7	73.6
SALE-MLP(Node2vec)	**85.6**	**75.1**
SALE-MLP(LINE)	83.2	71.5

As observed in table 4, SALE-MLP is consistently above the teacher SAGE irrespective of loss used. Further, Node2Vec performs the best due to the use of sophisticated and generalized representation formulation of Node2Vec. This also helps establish the fact that SALE-MLP outperforms the teacher due to its combine knowledge from GNN teachers and topological knowledge from structural loss.

4.2 Task Agnostic

We have also performed the task of link prediction as illustrated in Table 5

Table 5: Link Prediction AUC and accuracy in Inductive Settings

Model	Cora	Citeseer
Teacher	0.945/88.5%	0.947/88.1%
MLP	78/61.7%	0.88/78.9%
GLNN	0.88/81.2%	0.92/84.5%
NOSMOG	0.93/86.9%	0.90/80.0%
SALE-MLP **(Ours)**	**0.96/90.9%**	**0.98/94.8%**

To achieve this we picked the pre-trained model of node prediction. Subsequently, we extract the embedding for each node from the first layer and use this embedding to train another downstream MLP for the link prediction task. The downstream MLP takes the embedding of two nodes as input and outputs the probability of a link between them. For training, 80% data of the links are used, while testing is done on the remaining 20%. To compare the performance we have reported both AUC and accuracy scores for teacher and distillation methods. As observed from the reported performance SALE-MLP is superior in terms of both AUC and accuracy, this substantiates the usefulness of the proposed approach in the Link prediction task.

4.3 Teacher Agnostic

Moreover, we have also evaluated SALE-MLP with other SOTA methods on the node classification in inductive settings with different teacher GNNs: GAT[Veličković *et al.*, 2018] and GCN[Kipf and Welling, 2022]. The accuracy is reported in Table 6 with **bold** representing the best accuracy while underlined represents the second-best results.

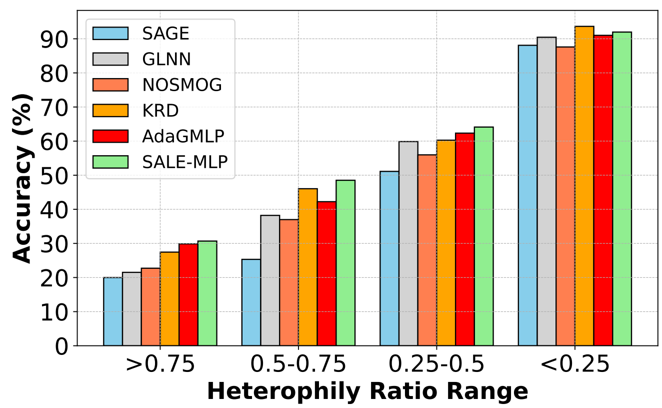
Table 6: Node Classification accuracy in Inductive Settings for different Teachers.

Method	Teacher	Cora	Citeseer	Pubmed	A-photo	A-computer
Teacher	GAT	83.0	70.6	80.5	82.4	90.5
MLP		59.4	58.3	68.4	67.6	73.0
GLNN		73.5	69.1	79.9	82.4	88.8
NOSMOG (w/o deepwalk)		74.7	69.9	80.0	80.7	92.2
NOSMOG (not Graph-free)		81.0	70.2	80.7	82.9	93.0
KRD		73.4	71.2	82.6	81.9	90.5
AdaGMLP		76.2	71.9	82.2	81.4	92.2
SALE-MLP **(Ours)**		**85.4**	**73.7**	**83.1**	**83.1**	**93.5**
Teacher	GCN	82.6	70.3	76.2	83.8	91.5
MLP		59.4	58.3	68.4	67.6	77.2
GLNN		75.6	73.5	77.5	80.8	89.7
NOMSOG(w/o deepwalk)		72.1	69.0	76.7	83.1	91.5
NOMSOG (not Graph-free)		80.5	72.8	78.1	85.2	92.0
KRD		76.7	71.2	78.3	84.1	88.6
AdaGMLP		75.6	71.8	77.2	82.5	89.6
SALE-MLP **(Ours)**		**82.5**	**73.2**	**79.1**	**84.3**	**93.0**

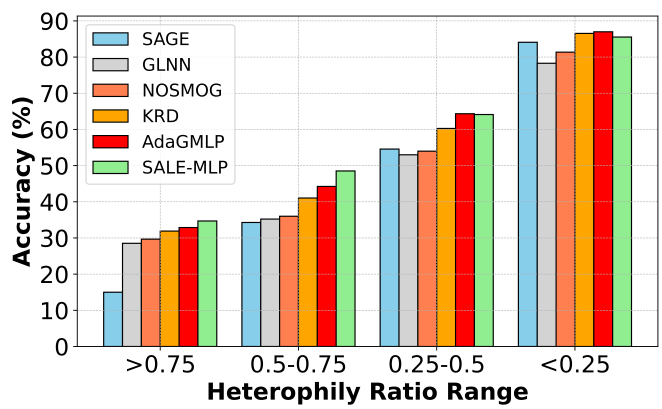
We chose inductive settings as this setting is more closely related to the real world where predictions are required on unseen nodes. The performance is consistent with what is observed in the main paper with SAGE[Hamilton *et al.*, 2017] as teacher, SALE-MLP outperforms the existing G2M methods while also outperforming teacher due to the varied information from both distillation and unsupervised structural loss. Further, SALE-MLP trained with GAT as teacher performs the best because the fact GAT itself achieves superior accuracy due to the attention. This establishes the teacher-agnostic nature of SALE-MLP where the proposed methodology can be applied to any teacher GNNs including attention networks.

4.4 Heterophily independent

We have also evaluated the SALE-MLP on real-world conditions characterized by high heterophily using the Cora and Citseer datasets along with the experiments in the main paper. As depicted in Figure 3 the heterophily ratio of nodes increases the performance tends to drop for all methods.



(a) Cora



(b) Citeseer

Figure 1: Performance of various G2M methods over different heterophily ratios

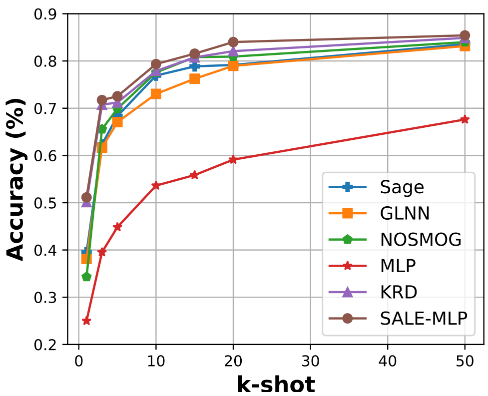
The Heterophily ratio is calculated as the percentage of neighboring nodes that have different labels than the source node. The plots signify that SALE-MLP performs slightly better than other G2M distillation approaches in case of high heterophily. The achieved performance is not drastically better because SALE-MLP doesn't directly include anything to counter heterophilic but the latent embedding in itself combines node features and structural information to some extent leading to a slightly better performance of the proposed SALE-MLP

5 Experimental Analysis

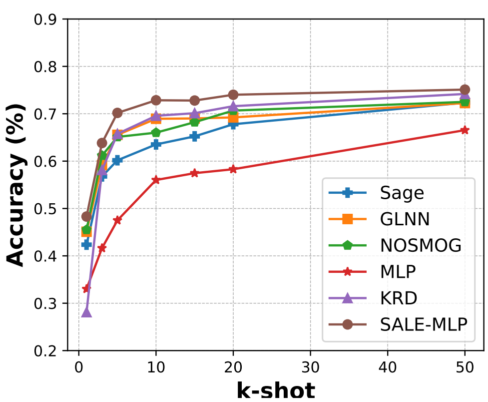
We have reported a few experimental results reported in the main paper here to have large/clear plots and visualization which were reduced in size in the main paper due to space constraints.

5.1 Generalization Ability of SALE-MLP

Evaluating generalizability is another aspect that is explored for different approaches in the Cora and Citeseer datasets. We trained and plotted the performance of the SALE-MLP and other G2M methods in limited labeled samples per class (k-shots) scenario, Figure 2 demonstrates SALE-MLP's superior generalization capabilities across different k-shot scenarios.

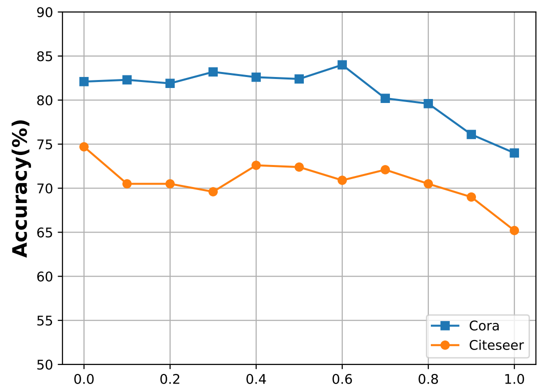


(a) Cora

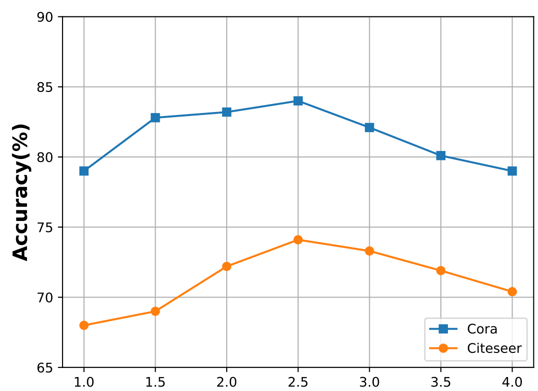


(b) Citeseer

Figure 2: Model accuracy vs k-shot plot for different datasets



(a) λ

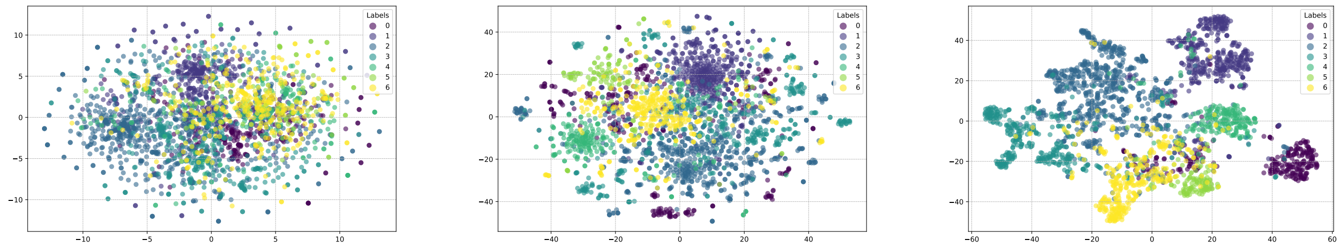


(b) α

Figure 3: Accuracy over different values of hyperparameters

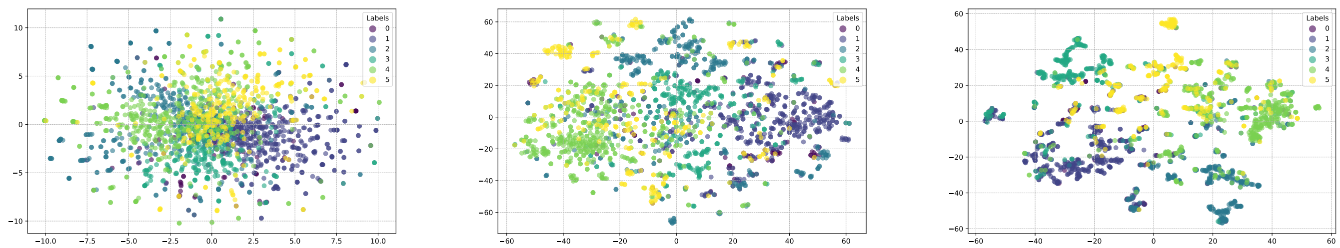
5.2 Hyperparameter Sensitivity Analysis

The impact of λ and α on SALE-MLP's performance using Cora and Citeseer datasets is examined in: Figure 3(a) showing model sensitivity to λ ([0,1] with step of 0.1) and Figure 3(b) illustrating sensitivity to α ([1,4] with step of 0.5).



(a) Node features (b) NOSMOG (c) SALE-MLP

Figure 4: t-SNE plots of the input feature space for Cora



(a) Node features (b) NOSMOG (c) SALE-MLP

Figure 5: t-SNE plots of the input feature space for Citeseer

5.3 Input Feature-Space Analysis

To understand the effectiveness of the generated latent space, We analyze the effectiveness of different feature spaces through t-SNE visualization. Figures 4(a) and 5(a) show the node-feature vs. node label plots, representing the input space used by most G2M methods (GLNN, KRD, AdaGMLP) while Figures 4(b) and 5(b) demonstrate NOSMOG's enhanced input space, incorporating DeepWalk embeddings. Finally, Figures 4(c) and 5(c) reveal SALE-MLP's latent space.

5.4 Statistically analysis

We performed a paired McNemar's Test to assess if the performance of SALE-MLP is statistically significant against the second-best performing model. Specifically, it tests the null hypothesis that the two models have equal proportions of correct predictions, focusing on instances where the classifiers disagree. As observed in Tab. 7 below, the p-value of SALE-MLP and the second-best distilled model for every dataset is less than 0.05, showcasing its statistical significance.

Table 7: P-value for McNemar's statistical test in inductive scenario.

Dataset	**Cora**	**Citeseer**	**Pubmeed**
SALE-MLP	0.01	0.0025	0.002

References

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